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Abstract

Table mining is a popular research field that involves complicated technologies, including information retrieval, data mining, visual and textual understanding and logical reasoning. With the emergence of Large Language Models (LLMs), the field has witnessed considerable advancements, presenting new paradigms for table understanding, extraction, and reasoning. In this survey, we conduct a comprehensive review of the literature on table mining with LLMs. We begin by introducing the fundamental overview of tabular data and possible challenges in LLM-based table mining. Specifically, we explore the challenges unique to this domain, such as heterogeneous table structures, contextual ambiguity, and domain-specific knowledge requirements. Then, we summarize representative tabular tasks in table preparation and mining, categorizing existing methods along dimensions including task scope, model architecture, and application scenarios. Next, we describe advanced LLM-based learning strategies in table mining, including foundation models and training-free methods. As specific issues, we review studies of trustworthy LLM-based table mining and some domain-specific applications. Finally, we discuss prospects and future directions in the field of LLM-based table reasoning, including issues of generalization, interpretability, efficiency, etc. We hope this survey can present valuable resources for researchers and practitioners, paving the way for further exploration in this field. The repository is at: <https://github.com/USTCAGI/Awesome-LLM-Table-Mining>.

A Survey on Table Mining with Large Language Models: Challenges, Advancements and Prospects

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Abstract—Table mining is a popular research field that involves complicated technologies, including information retrieval, data mining, visual and textual understanding and logical reasoning. With the emergence of Large Language Models (LLMs), the field has witnessed considerable advancements, presenting new paradigms for table understanding, extraction, and reasoning. In this survey, we conduct a comprehensive review of the literature on table mining with LLMs. We begin by introducing the fundamental overview of tabular data and possible challenges in LLM-based table mining. Specifically, we explore the challenges unique to this domain, such as heterogeneous table structures, contextual ambiguity, and domain-specific knowledge requirements. Then, we summarize representative tabular tasks in table preparation and mining, categorizing existing methods along dimensions including task scope, model architecture, and application scenarios. Next, we describe advanced LLM-based learning strategies in table mining, including foundation models and training-free methods. As specific issues, we review studies of trustworthy LLM-based table mining and some domain-specific applications. Finally, we discuss prospects and future directions in the field of LLM-based table reasoning, including issues of generalization, interpretability, efficiency, *etc.* We hope this survey can present valuable resources for researchers and practitioners, paving the way for further exploration in this field. The repository is at: <https://github.com/USTCAGI/Awesome-LLM-Table-Mining>.

Index Terms—Table Mining, Large Language Model, Table Understanding, Table Reasoning.

I. INTRODUCTION

TABLES are a critical medium for information dissemination. Due to the structured formats and semantic conciseness, tables have been widely adopted to present core information in documents, reports and websites precisely. As the exponential growth of tabular data carries out a rising trend of automated table mining, sophisticated frameworks integrating advanced artificial intelligence technologies have been proposed to target complicated table mining processes, involving retrieval [1; 2], logical reasoning [3], predictive analysis [4], and summarization [5]. Advancements in deep neural architectures and natural language processing have catalyzed innovative methodologies for interpreting complex table structures. Despite their success, effectively understanding the heterogeneous structures, mining contextual dependencies,

and integrating the entity knowledge still remain challenging in table mining.

Recently, *Large Language Models* (LLMs) have revolutionized the field of table mining by introducing brand new learning paradigms [6], which facilitate enhanced contextual comprehension, advanced logical reasoning and optimized knowledge representation. State-of-the-art LLM architectures such as GPT-series models have demonstrated remarkable performance in natural language processing (NLP) tasks. They subsequently pioneer novel methodologies for: (i) structural analysis of tables, (ii) semantic interpretation of table cell content, (iii) logical and computational table reasoning and (iv) cross-modal knowledge integration. Despite these advancements, a comprehensive review of the challenges, advancements, technologies, applications and prospects in table mining remains underexplored.

In this survey, we aim to provide a comprehensive review of the research efforts in table mining with LLMs. Specifically, we address the following key aspects: (1) We categorize and analyze the unique challenges in table mining, such as structural variability, contextual ambiguity, and task-specific requirements. (2) We review existing approaches and advancements in applying LLMs to table-related tasks and discuss innovations in contextual representation, knowledge integration, and reasoning capabilities. (3) We provide a taxonomy to classify existing work, focusing on task scope, model architectures, and applications. (4) We identify open challenges and future research directions, including improving model interpretability, enhancing computational efficiency, and expanding multi-modal integration. By addressing the aforementioned critical issue, this survey provides a comprehensive view of the broader challenges, advancements, and application scenarios in the field of table mining.

Our review is structured as follows: First, we present a fundamental overview of tables in Sec. III and then analyze the advantages and challenges of the integration of LLMs and table mining in Sec. IV. Sec. V focuses on the preparation processes before a wide range of table mining tasks described in Sec. VI. Then, Sec. VII describes advanced LLM-based foundation models and training-free methods. Sec. VIII presents trustworthy techniques integrated in table mining studies, while Sec. IX lists domain-specific table mining applications. Finally, Sec. X discusses open challenges and future directions. We hope this survey can present valuable resources for researchers and interested in table mining.

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II. RELATED WORK

A. History of Table Mining

Table mining has a rich history, driven by the increasing need for structured knowledge extraction from diverse data sources. Early research primarily relied on manual rule-based systems and statistical methods to process tabular data [7; 8]. These approaches were widely adopted in domains like finance and healthcare for structured data extraction, such as parsing financial ratios from HTML-based balance sheets in corporate filings, extracting medication dosages from electronic health records (EHRs) with nested table structures, and recovering gene-expression data tables from PDF-based biomedical literature through layout-aware parsing rules. While effective in well-defined, static environments and offering strong interpretability, these approaches were highly domain-dependent and faced scalability and adaptability challenges when handling heterogeneous, large-scale tabular datasets. The rise of machine learning injected new vitality into table mining. Tree-based models [9] and ensemble learning frameworks like Random Forests and Gradient Boosting Machines [10; 11] demonstrated enhanced generalization capabilities by automatically learning feature interactions. Despite these advancements, these methods required substantial manual feature engineering and domain expertise, such as the manual design of statistical or temporal features and handling of categorical features. The advent of deep learning marked a paradigm shift in tabular data mining. Architectures like Multilayer Perceptrons (MLPs) and autoencoders attempted to bypass laborious feature engineering through end-to-end learning of nonlinear relationships in tabular data [12]. However, their performance was constrained by the lack of task-specific architectures and susceptibility to overfitting. Recent breakthroughs in representation learning, particularly Transformer-based architectures, have significantly advanced the field. Models employing self-attention mechanisms, such as TabTransformer [13] and TabNet [14], utilize categorical feature embeddings and stacked attention layers to model long-range feature dependencies and dynamic interactions, showing promising results in various tasks.

B. Comparison with Existing Surveys

Numerous surveys have been conducted in the field of table mining. Earlier surveys introduce heuristics and hand-crafted patterns in the field of table recognition [7] and table processing [8]. While deep learning technologies have shown remarkable performance in tabular tasks, [15] focuses mainly on deep learning model architectures and pre-training objectives. Some conventional tabular tasks have been fully addressed in this period. [16] investigates the approaches of table detection and structure recognition integrating computer vision frameworks, while [17] reviews methods and applications of deep learning technologies in text-to-SQL parsing tasks. Nowadays, the rise of LLMs has promoted high potential in advanced and extended table mining tasks. [18] presents a refined survey of text-related table reasoning approaches integrating natural language processing methodologies, while [19] reviews LLM-based techniques in table prediction, generation and understanding. In comparison, [20] presents a relatively

thorough review of LLM-based table processing within advanced learning strategies. Some other surveys concentrate on specific domain applications (*e.g.*, scientific table recognition and analysis [21]) or single sources (*e.g.*, web table extraction [22]), without presenting a comprehensive review integrating advanced tasks, techniques and domain applications.

III. FUNDAMENTAL OVERVIEW

Tabular data have unique characteristics and diverse formats of expression, making them widely applied in various domains to convey critical information. This section presents a concise overview of the definitions and fundamental elements of tabular data, categorizes different table types and sources, highlights their characteristics associated with tabular data analysis, and summarizes related datasets and benchmarks.

A. Table Concept Description

Tables are a primary form to represent structured data [8], which organizes information through systematically arranged rows and columns. Generally, each *row* represents a record in the table, while each *column* corresponds to an attribute of the tabular data. Table *cells* serve as the basic building block associated with specific attribute columns for the record rows. These cell values can take various forms such as numerical data, texts or images. In most cases, tables are accompanied by *metadata* (*e.g.*, title, annotations, schema definitions) that provide additional semantic context and facilitate a better understanding of the table structure and content. An example of tabular data with descriptions of the basic concepts is illustrated in Table 1. Leveraging their (semi-)structured format, tables effectively integrate information from multiple sources, organizing complex data into an intuitive and comprehensible manner [22].

B. Table Types and Sources

Tabular data is generally recognized as a (semi-)structured data form [15; 20]. However, as tabular data has been increasingly applied across diverse domains, its sources and formats often vary depending on the characteristics of specific scenarios. Consequently, not all tabular data conforms to the (semi-)structured storage type. To address this variability, we categorize tabular data into three types: structured tables, semi-structured tables and unstructured tables. Furthermore, we outline several potential sources for each table type, as illustrated in Fig. 2.

1) *Structured Tables*: Structured tables adhere to a strict and standardized schema definition, in which all records and attributes conform to the pre-defined table schema and formats. This consistency enables efficient data organization and retrieval, making structured tables highly reliable for systematic analysis. In particular, *relational databases* (RDB) [23; 24; 25] are a prominent example of structured tables.

RDB Tables. RDB tables operate with fixed schema definitions and ensure data integrity by following explicit type specifications and key value constraints (*e.g.* primary keys and foreign keys). This implies that table operations such as

Paris 2024 Olympic Medal Table ④					
Rank	NOCs	Gold	Silver	Bronze	Total
1	United States of America	40	44	42	126
2	People's Republic of China	40	27	24	91
3	Japan	20	12	13	45
4	Australia	18	19	16	53
5	France	16	26	22 ③	64
...	...	...	...	...	...
Last updated: August 12, 2024 at 04:56 (your local time) ⑤					

Basic Table Concepts

① Row

② Column

③ Cell

④ Metadata (Title)

⑤ Metadata (Annotation)

Fig. 1. An example of tabular data with descriptions of basic table concepts: row, column, cell and metadata.

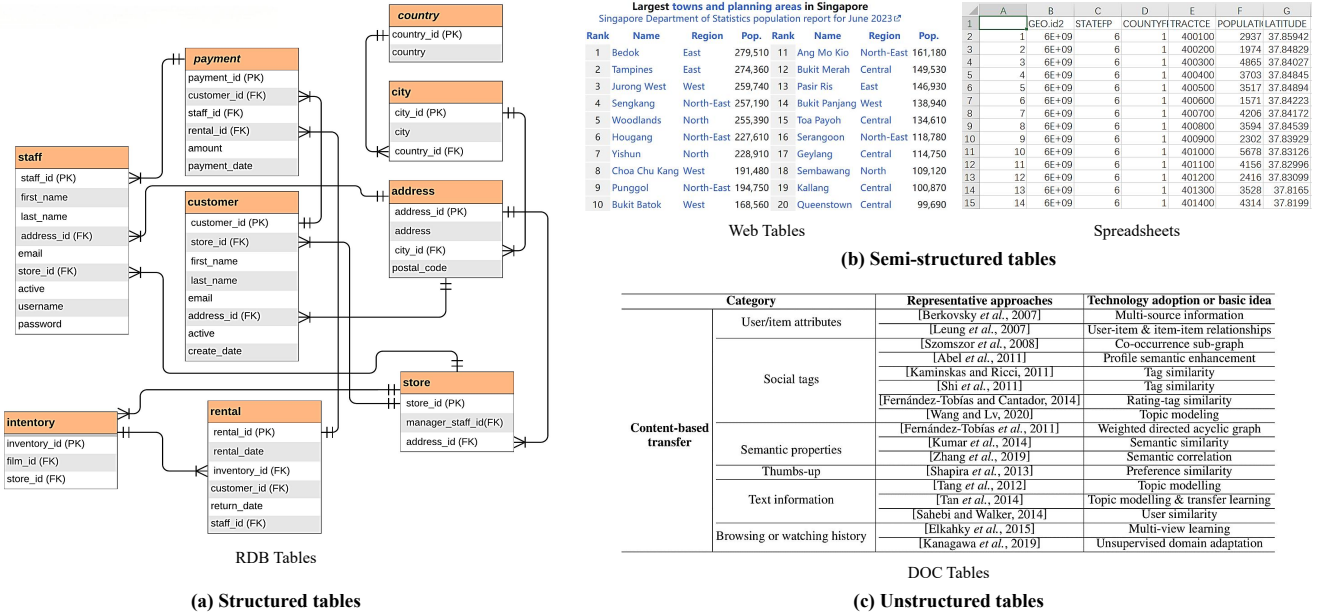


Fig. 2. Illustration of three types of tables: (a) structured tables, including relational database (RDB) tables, (b) semi-structured tables, including web tables and spreadsheets, and (c) unstructured tables, including document (DOC) tables.

inserting, deleting and updating must not only maintain the validity but also uphold the consistency and legitimacy of the affected tables. For instance, when entering a course's grade record into a student grade database, it is essential to ensure that each attribute's data type conforms to the schema definition. Additionally, foreign key constraints (*e.g.*, verifying the existence of the student's name and course name in their respective tables) must be satisfied to preserve referential integrity. In practical implementation, users interact with RDB using *structured query language* (SQL), which facilitates querying desired information and performing specific operations on database tables. RDBs support transactions that comply with the ACID principle (*i.e.*, atomicity, consistency, isolation, and durability), ensuring data integrity and avoiding unpredictable errors caused by transaction processing failures. As a result, RDB tables exhibit highly structured characteristics, prioritizing strict adherence to schemas and structural consistency.

2) *Semi-structured Tables*: In contrast, semi-structured tables lack strict schema definitions, allowing cell contents to vary in types, formats and modalities. Such flexibility makes semi-structured tables more adaptable and extensible in representing diverse information. Web tables [26; 27; 28] and spreadsheets [29; 30; 31] are considered important sources of semi-structured tables, which enable dynamic and heterogeneous content representation without being constrained by strict schema definitions.

Web Tables. Websites host a vast array of tabular data in various formats (*e.g.*, HTML, JSON, Markdown) and structures (*e.g.*, standard tables, hierarchical tables, multiple interconnected tables), as well as some auxiliary information (*e.g.*, annotation texts, attached charts). For instance, during the Olympic Games, online news platforms frequently summarize real-time medal standings in tabular form. These tables include associated subheadings, annotations (*e.g.*, update time, top-k list), visual elements (*e.g.*, national flags, medals) and other

enhancements to provide a more engaging and intuitive reading experience for audiences. Compared to the strict schema definitions in RDB tables, web tables pay more attention to the effective presentation of table content, emphasizing the communication of key information in an accessible and visually appealing way. Additionally, their semi-structured nature enables data analysts to efficiently extract, decode and pre-process various web tables using tools and technologies (e.g., web crawlers, programming languages). This adaptability makes web tables a valuable resource for downstream analysis and mining tasks.

Spreadsheets. Currently, the majority of workers employ spreadsheet software systems (e.g., Microsoft Excel¹, Google Sheets², WPS Spreadsheet³) and cloud-based spreadsheet documents to generate a substantial volume of spreadsheet tables. These systems are highly flexible and professional, allowing users to customize table structures by adjusting sizes, boundaries and merged cells. They also offer built-in operations (e.g., sorting, filtering, average calculation) to facilitate data analysis. Processed table data can be effectively visualized through charts and appropriately positioned annotations. Based on these advanced functionalities, a wide range of professionals use spreadsheet systems to handle domain-specific statistical data, produce financial table reports, and maintain essential personal information in memos. Similar to web tables, spreadsheets do not enforce rigid schema definitions but instead emphasize the data content itself. Moreover, spreadsheets rely on standardized file formats (e.g., .xlsx format) that are easy to read and save, allowing users to perform mining with only a few read preprocessing operations. In summary, spreadsheets are expected to retain a crucial role in future table-based research and applications due to their versatility and ease of use.

3) *Unstructured Tables:* The aforementioned tabular data exhibit distinct structured characteristics, which allows direct extraction of table contents for further analysis. However, in real-world applications, most tables are in an unstructured type. These tables are commonly found in articles, books and materials in image or PDF format, making it challenging to directly extract their digital information. *Document* (DOC) [7; 32; 33] tables are the most representative unstructured tables with complex layouts and contextual dependencies within surrounding texts.

DOC Tables. In scientific documents, including undigitized paper books, electronic files and instructional materials, tables are often embedded within documents in formats that need to be processed as image type data (e.g., .jpg format, .pdf format, .docx format). As a result, they cannot be directly stored or analyzed as formatted data types. For instance, tables used to display experimental results in scientific papers or personal information tables in Microsoft Word documents necessitate additional recognition and conversion processes. These steps transform the unstructured tables into native table formats that preserve clear corresponding structure and content

relationships before subsequent analysis and processing. To effectively extract and analyze these tabular data, additional preprocessing procedures are essential. For example, *document layout analysis* (DLA) [34; 35] may first be adopted to detect the DOC table objects. Then, visual multi-modal technologies can be employed to recognize and process unstructured tables from massive collections of documents. These analyses are of great significance in scientific research and technological applications.

C. Table Characteristics

Based on the definition and the diverse representation forms of tables, several distinctive characteristics of tabular data can be identified, which play a critical role in determining the inherent complexity associated with the processing, mining and analysis of tabular data. A list of table characteristics from the perspective of structure, content and relationship can be summarized below.

Structured Form. Unlike natural language text, tables are typically presented in a two-dimensional format with interwoven rows and columns, where each row and column can be regarded as a whole unit. This organization enables efficient indexing, retrieval and analysis in a logical arrangement.

Varied Boundaries. Tabular data in unstructured image formats present different kinds of boundaries (e.g., full-lined, partial-lined or unlined), which can influence the recognition of both structural layouts and content organization.

High-dimension Features. Tables often encapsulate a large amount of features. As the number of table features increases, the complexity of the data and the sophistication required for its analysis markedly escalate, necessitating advanced analytical techniques to process the tabular data effectively.

Multi-modal Content. Tables integrate content in diverse modalities, including texts, numerical values, categorical data, timestamps and even images. This heterogeneity poses challenges for further data understanding and processing.

Majority of Numerical Values. In most scenarios, tables integrate a majority of numerical values (e.g., experimental result tables, financial report tables). The processing and further analysis of tabular data inevitably integrate complex numerical computation.

Semantic Conciseness. Tables are generally utilized to convey highly concentrated information, presenting critical metrics and statistical insights in a compact framework. While this facilitates efficient comprehension, it also results in a high information density demanding careful interpretation.

Complex Relationships. The structural organization of tables and the relationships among different attributes maintain rich dependencies through the primary key (i.e., the unique identifier for a row record in an RDB table) and foreign key (i.e., the column(s) that establish a relationship between two tables). These relationships are crucial for comprehending the underlying semantics, facilitating advanced querying.

Multiple Table Types. Tabular data can originate from numerous data sources, including databases, spreadsheets, web tables and documents. Specific normalization for tables from different sources is required to address variations in structure and format before further analysis.

¹<https://www.microsoft.com/microsoft-365/excel>

²<https://workspace.google.com/products/sheets/>

³<https://www.wps.com/office/spreadsheet/>

TABLE I
REPRESENTATIVE DATASETS AND BENCHMARKS INVOLVING TABULAR DATA FOR FURTHER UNDERSTANDING AND MINING TASKS.

Task	Dataset	Source	Year	Statistics	Link
Table Recognition	TableBank [36]	ArXiv	2019	145.4k image tables	Link
	SciTSR [37]	ArXiv	2019	15k image tables	Link
	PubTabNet [38]	PubMed	2020	568k image tables	Link
	FinTabNet [39]	Business reports	2021	38.2k image tables	Link
	PubTables-1M [40]	PubMed	2021	947.6k image tables	Link
	WTW [41]	Wild Scenarios	2021	14.5k image tables	Link
	TabRecSet [42]	Wild Scenarios	2023	38.1k image tables	Link
Table QA	WikiTQ [43]	Wikipedia	2015	2.1k tables, 22k questions	Link
	WikiSQL [44]	Wikipedia	2017	24.2k tables, 80k queries	Link
	HybridQA [45]	Wikipedia	2020	13k tables, 70k questions	Link
	FinQA [46]	Financial reports	2021	2.7k report pages, 8.2k questions	Link
	TAT-QA [47]	Financial reports	2021	2.7k hybrid contexts, 16.5k questions	Link
	HiTab [48]	StatCan, NSF, etc.	2021	3.5k tables, 10.6k questions	Link
	FeTaQA [49]	ToTTo	2022	10.3k tables, 10.3k questions	Link
	MultiHiertt [50]	FinTabNet	2022	2.5k documents, 10.4k questions	Link
	MMTabQA [51]	WikiTQ, etc.	2024	25k tables, 69.7k questions	Link
Table Fact Verification	TabFact [3]	Wikipedia	2020	16.5k tables, 117.8k statements	Link
	FEVEROUS [52]	Wikipedia	2021	28.7k tables, 87k statements	Link
	SciTab [53]	SciGen	2023	199 tables, 1.2k statements	Link
Text-to-SQL	WikiSQL [44]	Wikipedia	2017	24.2k tables, 80k queries	Link
	Spider [54]	WikiSQL, etc.	2018	200 DBs, 10.1k questions	Link
	BIRD [55]	Kaggle, etc.	2023	95DBs, 12.7k queries	Link
Table-to-Text	ToTTo [56]	Wikipedia	2020	83.1k tables, 131.8k examples	Link
	HiTab [48]	StatCan, NSF, etc.	2021	3.5k tables, 10.6k questions	Link
Table Entity Linking	S2abEL [57]	PapersWithCode	2023	732 tables, 8.4k annotations	Link
Column Type Annotation	GitTables [58]	GitHub	2023	1m tables	Link
Spreadsheet Manipulation	SpreadsheetBench [59]	Excel forums	2024	912 instructions, 2.7k samples	Link
Data Analysis	DAEval [60]	GitHub	2024	124 files, 603 questions	Link

D. Datasets and Benchmarks

As tabular data are widely utilized across various domains, there have been numerous high-quality, large-scale datasets and benchmarks for tasks related to the preparation, understanding and analysis of tabular data. Table I presents a comprehensive list of representative datasets and benchmarks for mining tasks involving tabular data. Traditional tasks include table recognition, table question answering (QA), text-to-SQL, table fact verification and table-to-text generation, each with a wealth of established datasets and benchmarks. Recent studies have shifted focus towards more complex and challenging tasks, such as table entity linking, column type annotation, spreadsheet manipulation and automatic data analysis, accompanied by some newly proposed datasets and benchmarks. Description, approaches and discussions of these tabular tasks will be presented in Sec. V and Sec. VI.

IV. ADVANTAGES AND CHALLENGES IN LLM-BASED TABLE MINING

In recent years, the rise of *Large Language Models* (LLMs) has significantly advanced the field of table understanding and mining, demonstrating great potential due to their powerful capabilities. This section provides a comprehensive overview of the advantages and challenges associated with LLM-based table mining procedures.

A. Advantages

LLMs have demonstrated considerable potential in the field of table mining. They leverage natural language interfaces

to provide users with efficient and intuitive human-computer interaction. With their extensive pre-trained knowledge and strong language comprehension capabilities, LLMs can tackle complex reasoning tasks in a variety of scenarios. Overall, LLMs possess several key advantages when processing tables in mining tasks, which can be summarized below.

Extensive Zero-shot Generalization. LLMs demonstrate remarkable zero-shot generalization ability, which arises from extensive pre-training on large-scale datasets across various domains. Given the tabular data, LLMs can easily capture its structural and semantic information through their intrinsic linguistic patterns and factual knowledge without further fine-tuning, which provides a brand new paradigm to complete table mining tasks.

Novel Generation and Interaction Paradigm. LLMs adopt the paradigm of textual generation and interaction rather than conventional knowledge representation learning, showing high flexibility and explainability in processing with reasoning tasks. In table understanding and reasoning procedures, LLMs can flexibly decompose the task objectives into several sub-steps and provide a transparent and comprehensive depiction of their intermediate reasoning process.

Rich World Knowledge. LLMs incorporate an extensive collection of multi-modal data from various domains during the pre-training stage. This enables LLMs to store and understand a wide range of knowledge, so that they are widely utilized to perform contextual understanding and reasoning on tables directly or after a short period of fine-tuning for knowledge transfer on domain-specific tabular data.

Strong Reasoning Ability. LLMs possess remarkable rea-

Model	FB15k-237-IMG				WN18-IMG			
	MR	Hits@1	Hits@3	Hits@10	MR	Hits@1	Hits@3	Hits@10
<i>Non-Transformer KGR Models</i>								
TransE	323	19.8	37.6	44.1	357	4.0	74.5	92.3
DisMult	512	19.9	30.1	44.6	665	33.5	87.6	94.0
ComplEx	546	19.4	29.7	45.0	-	93.6	94.5	94.7
ConvE	249	22.5	34.1	49.7	-	41.9	47.0	53.1
RGCN	600	10.0	18.1	30.0	-	8.0	13.7	20.7
IKRL(UNION)	298	19.4	28.4	45.8	596	12.7	79.6	92.8
TransAE	431	19.9	31.7	46.3	352	32.3	83.5	93.4
RSME(ViT-B/32+Forget)	417	24.2	34.4	46.7	-	94.3	95.1	-
<i>Transformer KGR Models</i>								
KG-BERT	153	-	-	42.0	58	11.7	68.9	92.6
VisualBERT	592	21.7	32.4	43.9	122	17.9	43.7	65.4
ViLBERT	483	23.3	33.5	45.7	131	22.3	55.2	76.1
MKGformer	252	<u>24.3</u>	<u>36.0</u>	<u>49.9</u>	25	<u>93.5</u>	<u>95.8</u>	<u>97.0</u>
SGMPT (Ours)	<u>238</u>	25.2	37.0	51.0	<u>29</u>	94.3	96.6	97.8

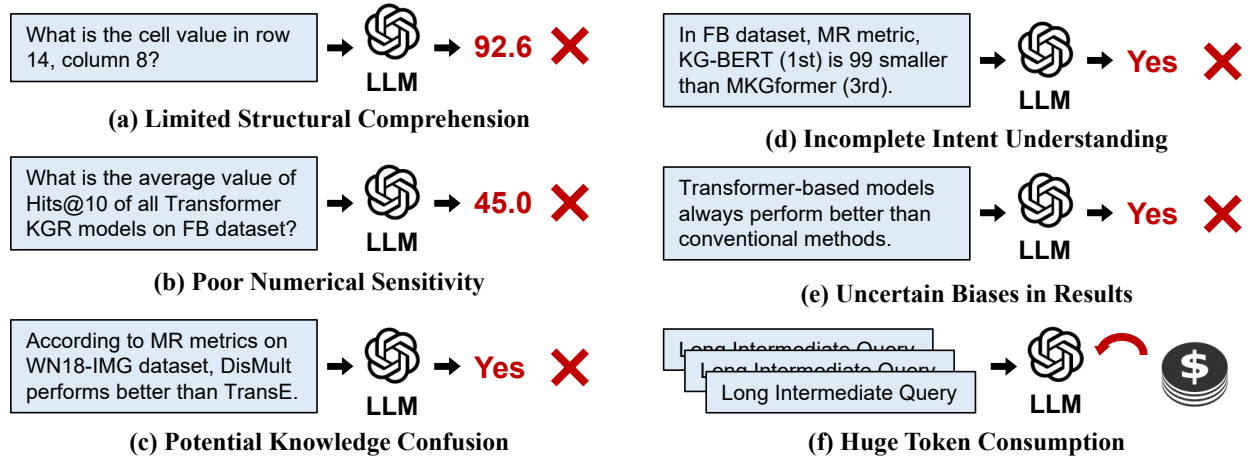


Fig. 3. Potential challenges in LLM-based table understanding and analysis.

soning abilities as a result of massive training and instruct tuning. Leveraging advanced capabilities of task planning, self-reflection, and tool integration, they can execute intermediate steps more accurately along the entire reasoning pathway. This empowers LLMs with great potential to perform table-based multi-step reasoning and compositional reasoning.

B. Challenges

The emergence and development of LLMs have revolutionized the research field of table understanding and mining. However, alongside these remarkable advancements, a range of new challenging issues have been raised from the extensive explorations and applications, which may cause unreliable results with performance degradation. The challenges in LLM-based table analysis are shown in Fig. 3, summarized below:

Limited Structural Comprehension. While LLMs can handle a certain degree of structural information after training on massive textual corpus, they may struggle with parsing tables with highly complex structures. This limitation arises partly from the one-dimensional text-form encoding of two-dimensional tabular data (especially with multiple hierarchical indexes, merging cells), leading to comprehension gaps.

Poor Numerical Sensitivity. While LLMs exhibit strong capabilities in understanding discrete textual information, they

demonstrate insensitivity to continuous numerical data, limiting their ability to handle table mining tasks involving extensive numerical computation and logical reasoning. Direct queries to LLMs may result in incorrect intermediate computational steps, ultimately leading to erroneous results.

Potential Knowledge Confusion. Tables present highly condensed information in rows, columns and cell contents, often comprising specialized terminology, concise symbols, or semantically ambiguous entities. As LLMs may suffer from a lack of domain-specific knowledge, they may produce erroneous intermediate results with irrelevant facts or fabricated contents, misleading the entire table mining procedure.

Incomplete Intent Understanding. LLMs may encounter limited performance in implicit intent understanding. In most cases, table reasoning involves multiple sub-facts to be verified or executed, which can be easily ignored in the LLM-based reasoning chain and also in consistency checking of the overall process.

Uncertain Biases in Results. LLMs may inherit biases from their pre-training data that can influence their reasoning performance. When the information within tabular tasks conflicts with models' biases, the output may become distorted, reducing the reliability and accuracy of the results.

Huge Token Consumption. When performing multi-turn

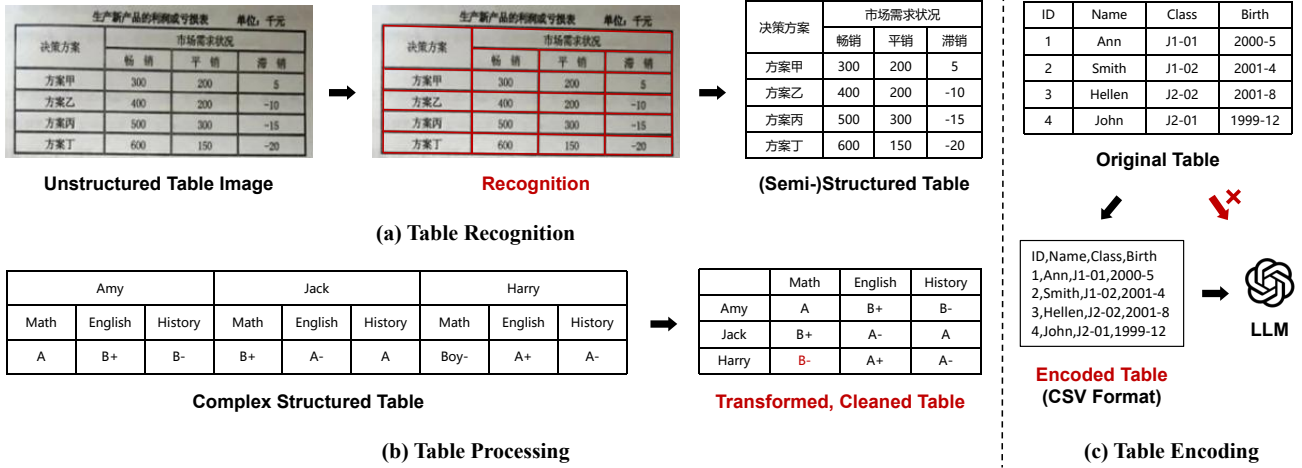


Fig. 4. An example of table recognition, table processing and table encoding involved in table preparation.

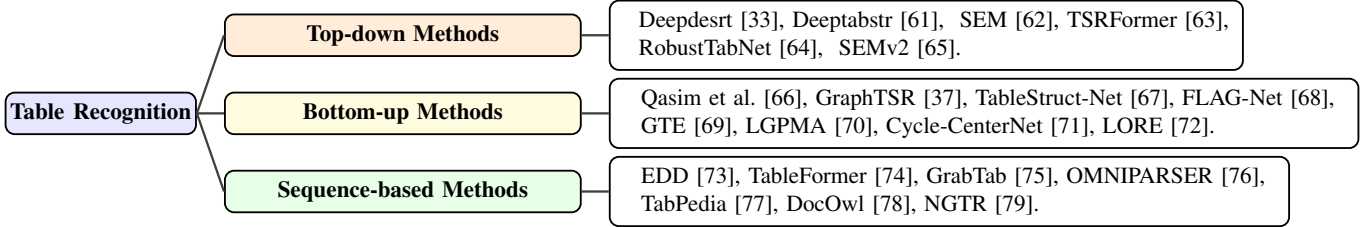


Fig. 5. Taxonomy and approaches of the table recognition task.

LLM reasoning over large tables, its long-context characteristic results in substantial token consumption. Therefore, it is crucial to explore cost-effective and efficient LLM-based table mining methods.

V. TABLE PREPARATION

Table preparation aims at transferring the unstructured, non-standard and unclean tables into structured, complete and standardly formatted tables for further LLM-based table mining. This section systematically explores the key tasks involved in table preparation, including table recognition, table processing, and table encoding, presenting a thorough overview of the table preparation process. Examples of the involved processes are shown in Fig. 4.

A. Table Recognition

Table recognition aims to convert unstructured tabular data into structured formats by identifying table cell content, determining cell relations and transforming them into structured formats. Related studies can be categorized into top-down, bottom-up, and sequence-based methods based on their strategies, as shown in Fig. 5.

Top-down Methods. Top-down approaches first predict table boundaries and then identify the locations of individual cells through the intersections of rows and columns, and finally merge any over-separated cells. DeepDesert [33] first applied the Faster R-CNN segmentation model to table structure recognition, identifying rows and columns through semantic segmentation and generating cell representations based on

their intersections. However, this method introduces heuristic masks during the post-segmentation processing, leading to poor generalization performance. DeepTabStr [61] proposed applying deformable convolution to row and column prediction to address this issue, directly generating row and column representations from table images to better adapt to different table structures. SEM [62] also builds on this framework by incorporating text information from the cells. By combining BERT and RoIAlign technologies, SEM directly extracts grid-level visual and textual features, fusing them to complete embedding and merging operations. Unlike traditional table row and column region recognition methods, TSRFormer [63], RobustTabNet [64], and SEMv2 [65] employ a strategy of directly regressing the row and column separators. This approach generates more accurate row and column boundaries, reduces sensitivity to cell alignment, and demonstrates stronger robustness when processing complex tables. As a result, top-down methods generate a grid structure by interweaving rows and columns, which ensures good generalization ability. However, their performance depends on the accuracy of column recognition and cell content extraction still requires an OCR model for subsequent processing.

Bottom-up Methods. Bottom-up approaches first recognize cells, then treat the detected cells as nodes in a table graph, and predict the logical relationships between the cells based on the constructed graph. [66] first proposed to use the bottom-up paradigm to solve the table recognition problem. This method first identifies the cell content through OCR technology and then establishes the relationship graph between cells through the graph neural network. GraphTSR [37] extracts cell content

and bounding boxes directly from PDF and then constructs an graph to predict the adjacency relationship between cells. FLAG-Net [68] uses transformer to aggregate cell context information to achieve table recognition without relying on document metadata. The text boundary-based method relies on pre-extracted cell information, which limits the generalization ability of the method. TabStruct-Net [67] first proposed jointly detecting cells and predicting their relationships. GTE [69] applies the target detection network to cell recognition and merges cells through GTE-Cell. However, this method has limitations in effectively identifying empty cells. LGPMA [70] detects the cell bounding box and uses the local branch (LPMA) and global branches (GPMA) to finely align the bounding box. Bounding box-based methods perform well on images with well-defined bounding boxes but perform poorly when faced with tabular images with geometric distortion or curved box lines. Cycle-CenterNet [71] simultaneously detects the vertices and center points of cells, learns the features of common vertices, and effectively groups cells, thereby improving the processing ability of curved tables. LORE [72] recognizes tables by detecting cell corners. It predicts the logical position and spatial position of table cells at one time, simplifying the structure.

Sequence-based Methods. The sequence-based method utilizes a serialized model for table recognition, commonly based on Transformer framework [80], where an image serves as the encoder input, and the decoder generates structured information of the table. Subsequently, traditional OCR tools or trained decoders are employed to extract the text content from each cell for structured integration. EDD [73] first proposed an end-to-end table recognition paradigm to directly convert table images into HTML code. This method adopts an encoder-decoder architecture, where one decoder reconstructs the structure of the table, and the other decoder recognizes the text content in the cells. TableFormer [74] extends this end-to-end approach by applying it to predict the bounding boxes of table cells. The cell content is directly extracted from PDF files, so this method cannot handle image input. GrabTab [75] introduces a new component deliberator capable of adapting to most complex table images without requiring complex post-processing. Recent advancements in table recognition have also moved towards generalizing models to handle multiple tasks. Rather than focusing on table recognition tasks, these methods explore the potential of unified solutions that can be applied across different domains [81]. By pre-training or fine-tuning LLMs on extensive, diverse datasets, such models can be adapted to various fields, including table recognition and visual question answering. OMNIPARSER [76] introduces a unified Transformer framework that performs multiple text-based parsing tasks end-to-end. This strategy effectively reduces the complexity of learning structured sequences, improving generation efficiency and accuracy. In contrast to traditional Transformer training methods, TabPedia [77] builds a unified framework by training a large visual-language model. Leveraging the capabilities of LLMs, this framework can simultaneously handle tasks such as table detection, table structure recognition, and table question-answering. To address the limitations of existing multimodal models in perceiving

text structure information, DocOwl [78] introduces a structural information encoder to enhance the model's ability to process structured data. NGTR [79] explores completing the table recognition task by employing vision LLMs in a training-free reasoning paradigm and proposes a framework integrating multiple lightweight models and a neighbor retrieval mechanism to improve recognition accuracy.

B. Table Processing

Table processing is a key step in table understanding, primarily involving table transformation and table cleaning. Proper table processing enables better extraction of latent information from tables, laying a foundation for subsequent tasks such as data extraction, table understanding, or table reasoning.

Table Transformation. To adjust and modify the structure, format, and presentation of table contents, tabular data needs to be transformed. [30; 82] introduce the TableProg language that describes table data transformation rules, and develop the ProgFromEx algorithm to automatically infer programs that meet the required transformations based on user-provided input-output examples. [83] focuses on converting non-relational tables into standard relational tables. This approach utilizes visual patterns of tables and a self-supervised learning framework to generate training data and automatically identify suitable transformation operations.

Table Cleaning. Tables derived from real-world data often contain erroneous, non-standard, or irrelevant data. This issue not only affects data quality but also reduces the reliability of analysis results, making table cleaning a necessary step. The core tasks of table cleaning include error detection and correction. Traditional methods rely on rules and integrity constraints [84], such as functional dependencies and denial constraints, to identify duplicates, missing values, and data violations. Statistical methods, such as ERACER [85], enhance the flexibility and accuracy of cleaning by modeling and inferring through relational dependency networks. In recent years, machine learning methods [86] have demonstrated strong cleaning capabilities, which impute missing values in heterogeneous data.

In the context of the new era, with the rapid development of LLMs, table processing technology has encountered new opportunities and challenges. Recent research [87] reveals that LLMs can learn rich language and knowledge representations through pretraining, enabling them to perform table cleaning, data imputation, error detection, and other tasks more efficiently. LLMs can achieve multimodal data fusion, adaptive reasoning, reinforcement learning, and enhanced reasoning, and are increasingly gaining attention in table processing tasks.

C. Table Encoding

LLMs accept sequential texts as inputs. When processing tabular data, the structured tables must first be converted to specific forms for LLM encoding. Specifically, tabular data can be encoded into several formats, including JSON, LaTeX, CSV, XML, DataFrame, Markdown and HTML. Examples of table encoding formats are shown in Fig. 6.

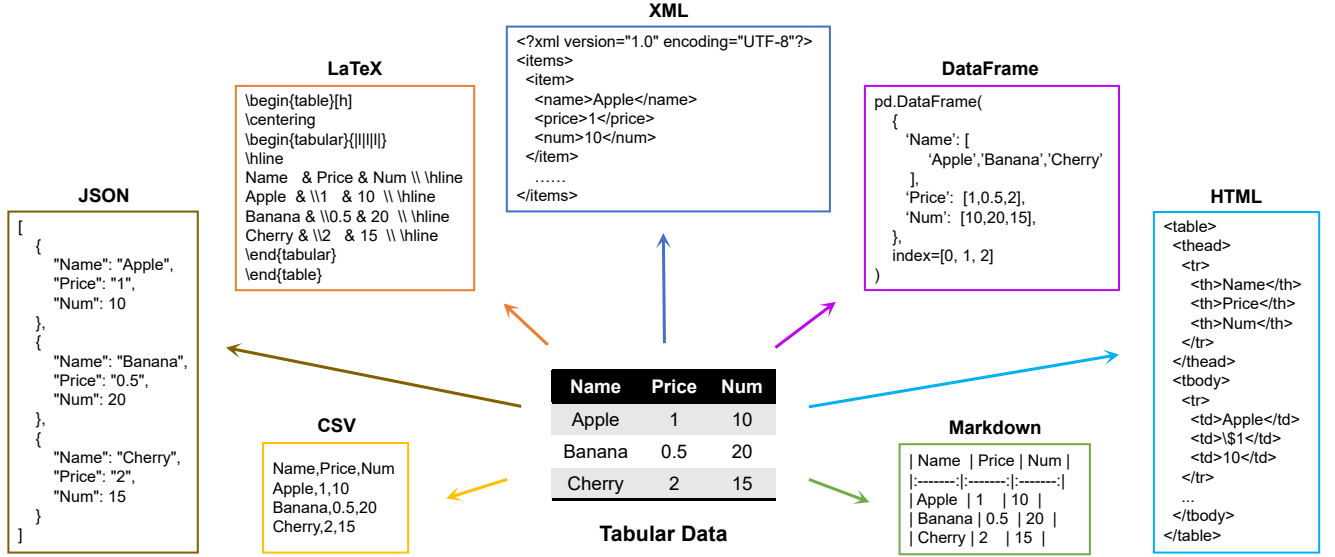


Fig. 6. Tables can be encoded into various formats, including JSON, LaTeX, CSV, XML, DFLoader, Markdown and HTML.

JSON. A lightweight data interchange format. While it is highly versatile, the nested objects can increase token usage when working with LLMs.

LaTeX. A high-quality typesetting system, offering extensive support for complex table representation through its rich package ecosystem. The complex syntax poses challenges for correct generation.

CSV (Comma-Separated Values). A simple file format using commas to separate values to store tabular data. Tables in this format are easy to read and write, yet may lack explicit coordinates and cannot represent complex structures.

XML. A flexible, extensible, structured markup language that supports complex data representation with nested elements, attributes and schema validation. However, its verbosity can also lead to inefficient token usage.

DataFrame. Python pandas dataframe is a versatile data structure commonly used for storing tabular data. While it simplifies data manipulation tasks, it requires careful configuration to handle different sources.

Markdown. A lightweight markup language with plain text formatting syntax. It offers a straightforward syntax and high readability, yet it cannot represent complex structures or accommodate additional attributes.

HTML. The standard language for creating and structuring content on the web. It can well support tables with complex hierarchical structures. However, its reliance on tags for structural representation makes it verbose and results in high token consumption when using LLMs.

Given the wide range of encoding methods available, selecting the most appropriate representation depends on several factors, including data complexity, model efficiency, token efficiency, and the specific use case. In practice, researchers must balance between representation fidelity with computational cost [88]. Moreover, combining multiple encoding methods can be advantageous in meeting different task requirements. For instance, LaTeX can be used for detailed representations,

complemented by Markdown for simplicity and readability, while CSV files can be augmented with supplementary metadata files (e.g., JSON sidecars) to encapsulate additional attributes.

VI. TABLE MINING

Table mining encompasses a wide range of tasks and applications (e.g., table question answering (QA), table imputation, table-to-text generation, table classification). In the era of LLMs, there have been significant advancements in the field of table mining research. This section provides a comprehensive overview of representative table mining tasks with related LLM-based approaches. An overview of the related definitions and descriptions is presented in Table II.

A. Table Question Answering

Table Question Answering (QA) aims to extract and process precise information from the given tables to answer the raised correlated questions. As a conventional NLP task involving precise processing of unstructured texts and structured tables simultaneously, table QA has garnered substantial research interest, especially in the era of LLMs bringing brand new learning strategies [89]. Existing LLM-based approaches primarily focus on three aspects of challenges: structures, contents and procedure. Taxonomy and approaches are shown in Fig. 7.

Structure Perspective. Numerous LLM-based studies have addressed the challenges of the various tabular data structures. When processing large-scale tables, encoding the whole tabular data into LLMs results in poor performance and high expenses, especially in mainstream multi-step reasoning procedures. Dater [90] adopts LLMs as versatile decomposers by selecting columns and rows that are relevant to the questions. TabSQLify [93] utilizes SQL queries to extract task-related small tables before further analysis. TAP4LLM [94] covers a table sampling module to decompose large tables into manageable sub-tables. ALTER [95] only encodes a small

TABLE II
AN OVERVIEW OF REPRESENTATIVE TABLE MINING TASKS WITH SHORT TEXTUAL DESCRIPTIONS.

Task	Description
Table QA (Sec.VI-A)	Given a question, extract evidence from the given tables to answer the question.
Table Prediction (Sec.VI-B)	Predict target values of a missing attribute or record in the tabular data.
Table Imputation (Sec.VI-B)	Complete the missing cell data in the tabular data.
Table Fact Verification (Sec.VI-B)	Given a statement, extract evidence from the given tables to verify if it is true.
Table Content Retrieval (Sec.VI-C)	Retrieve relevant information from tables based on natural language descriptions.
Table Classification (Sec.VI-D)	Classify the entire table or the individual columns with in a table.
Table-to-Text Generation (Sec.VI-E)	Generate natural language descriptions or summaries from the given tables.
Table Generation (Sec.VI-F)	Convert unstructured texts or other formatted data into organized tables.
Table Manipulation (Sec.VI-G)	Understand, process and generate table data using machine learning techniques.

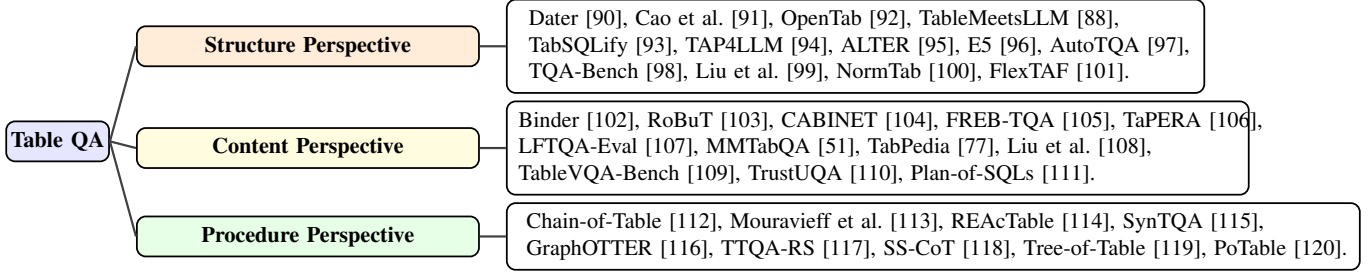


Fig. 7. Taxonomy and approaches of the table QA task.

subset of relevant tabular data and supplements it with a pre-augmented schema. In addition, it is not easy to understand non-standard tables (e.g., hierarchical tables) through direct LLM queries. [91] transforms received tables into a single hierarchical table and then represents it using a Pandas multi-index representation. E5 [96] introduces a code-augmented framework that encompasses self-explaining the hierarchical structures before further information extraction. Some scenarios include multiple tables with more complicated table-to-table relations. OpenTab [92] uses a retriever to retrieve relevant sampled tables from a corpus of tables for the query in open-domain table QA. AutoTQA [97] employs multi-agents with scheduling algorithms and collaboration across multiple tables. TQA-Bench [98] is a multi-table QA benchmark to evaluate the ability of LLMs to process diverse relational data. Partial studies are conducted to comprehensively evaluate the LLMs in structural understanding. TableMeetsLLM [88] designs a fine-grained benchmark including cell lookup, row retrieval and size detection to validate the structural understanding capabilities. [99] explores the robustness of LLMs to structural perturbations in tables and proposes a structure normalization method. NormTab [100] normalizes web tables to reduce the influence of structural variance and inconsistency to support symbolic reasoning. FLEXTAF [101] employs flexible tabular formats to enhance table understanding in different instances. As a result, these studies have well addressed the structure issue in table QA and achieved promising performance.

Content Perspective. Another type of study focuses on diverse table contents and related elements. Some approaches manage to bridge the unaligned table contents and query texts in LLM-based understanding. Binder [102] binds LLMs to extend the code grammar coverage through their knowledge for query-oriented table content operations. [108] integrates external information in a given table by constructing an augmenting table containing missing information for further query

generation. CABINET [104] suppresses extraneous information of tabular data beyond the question through a well-trained unsupervised relevance scorer. In addition, tabular data content can be presented in multi-modal form besides unstructured text elements. MMTabQA [51] explores the ability to reason on tables integrating both images and texts by proposing a new dataset. TabPedia [77] unifies visual table understanding tasks with concept synergy mechanism and proposes ComTQA as a visual table QA benchmark. TableVQA-Bench [109] directly collects image tables to establish a visual table QA benchmark to evaluate the capabilities of multi-modal LLMs. Some task datasets require long-form text answers to the question rather than short corpus or values. TaPERA [106] contains QA-based plans and then generates answer texts conditioned on the plan to reduce hallucination. LFTQA-Eval [107] rigorously assesses the efficacy of existing automated metrics in assessing long-form answers, raising improving issues over the metrics. Some other studies evaluate the robustness and explainability of table QA systems. RobuT [103] is an evaluation benchmark including human-annotated adversarial perturbations in terms of table header, table content and question. FREB-TQA [105] formalizes desiderata for a fine-grained evaluation of table QA systems, including reasoning dependence of table contents rather than biases. TrustUQA [110] is a trustful framework supporting multiple data types through condition graph as a unified knowledge representation method. POS [111] solves QA tasks through sole SQL executions to enhance the interpretability of generated answers. Overall, these studies have explored extensively to address the challenges related to content elements.

Procedure Perspective. Table QA often involves complex logical reasoning procedure that requires fine-grained design and employment. An intuitive idea is to decompose the reasoning procedure into multiple operations for step-by-step LLM queries. Chain-of-Table [112] pre-defines some atomic

operations for the LLM to dynamically plan the next executing operation. Tree-of-Table [119] employs table condensation and decomposition and constructs a hierarchical table tree that facilitates tree-structured reasoning. GraphOTTER [116] transforms the table into a graph representation and then establishes a step-by-step reasoning process on this graph. More advanced approaches consist of several role modules responsible for distinct purposes. ReAcTable [114] introduces the ReAct paradigm into the table QA task, integrating external tools like SQL and Python to enhance intermediate data representations and answer quality. TTQA-RS [117] contains a retriever and a reader that for table-text information retrieval and knowledge augmentation in reasoning-based QA. SS-CoT [118] instructs the LLM first to seek relevant information and then solve the questions that improve efficiency and reliability. PoTable [120] constitutes a planner and an executor with an LLM and a Python interpreter to coherently accomplish the task in a series of logical stages, enhancing answer accuracy and explainability. Other studies pay attention to the text-to-SQL paradigm as another choice for table QA reasoning. [113] imitates a restricted subset of SQL-like algebraic operations, providing intermediate supervision steps that allow for increased generalization and structural reasoning. SynTQA [115] identifies different strengths and weaknesses in text-to-SQL and end-to-end table QA paradigms and thus proposes to integrate them to exert their advantage simultaneously. These studies have achieved effective and understandable general reasoning.

To review the overall studies, current table QA systems have made considerable advancements in handling complex structures and understanding general semantic contents. Meanwhile, by utilizing advanced LLM reasoning strategies, existing studies can easily answer retrieval questions and also simple reasoning questions, with extensions in certain specialized contexts (*e.g.*, low-resourced languages [121], temporal reasoning [122]). However, there remains substantial room for further development in long-form subjective QA, domain-specific knowledge-based QA, and multi-table multi-hop questioning that demands more complicated, intricate logical reasoning, representing promising evolution directions for future research.

B. Table Reasoning

Table reasoning almost covers all downstream table analysis tasks. This subsection focuses on numerical reasoning and logical reasoning. The former emphasizes the prediction and inference of specific numerical values, including table prediction and imputation tasks. The latter pays more attention to analyzing and organizing the logical relationships inherent in factual information, in which table fact verification is a representative task.

Table Prediction. Table prediction involves predicting missing or target values in tabular data, which has been widely applied in specific domains (*e.g.*, implicit record recommendation [123; 124], time-series analysis [125; 126], exercising performance prediction [127; 128]). It can be divided into one-column prediction, focusing on predicting a specific column

using other columns, and whole-table prediction, which aims to fill missing values across the entire table [19]. Recent advances leverage LLMs to enhance prediction accuracy and generalization. One-column prediction focuses on predicting the values of a specific column based on other columns in the table, requiring careful handling of feature dependencies, avoiding data leakage, and ensuring the model captures the relationships between columns accurately. Some approaches use LLMs for prediction directly without fine-tuning, which we refer to as inference-only prediction. TABLET [129] utilizes models like TkInstruct to perform inference. LIFT [130] fine-tunes pre-trained language models such as GPT-3 using LoRA on the training dataset. TabLLM [131] uses T0 model and T-few for fine-tuning. TabLLM has demonstrated remarkable few-shot learning capabilities, outperforming traditional deep-learning methods and gradient-boosted trees. Whole-table prediction focuses on predicting missing values across the entire table, requiring careful modeling of row and column relationships, maintaining consistency across predictions, and ensuring the model effectively captures global patterns within the data. LLM-based methods in whole-table prediction could be classified into two types: inference-only prediction and fine-tuning [132]. For the former, ZeroTS [133] examines the use of LLMs like GPT-3 and LLaMA-70B directly for time series forecasting. For the latter, TimeLLM [134] presents a novel approach to time series forecasting by fine-tuning LLMs like LLaMa and GPT-2. It demonstrates notable performance in few-shot learning scenarios, where only 5 percent or 10 percent of the data are used.

Table Imputation. Before performing table reasoning, the content of the table must be processed. Proper table processing enables better extraction of latent information from tables, facilitating subsequent tasks. Among the various steps of table processing, table imputation is the most critical. Table imputation refers to estimating the missing data in a table using specific algorithms or models and filling in the missing values with reasonable approximations. Table imputation not only improves data quality but also enhances the accuracy and reliability of subsequent data analysis tasks. Currently, the techniques for table imputation can be broadly classified into statistical-based and machine learning-based methods. Statistical methods typically rely on simple statistics (such as mean, median) or regression analysis techniques to impute missing values, as exemplified by common interpolation methods [135]. In recent years, machine learning and deep learning methods have gradually become the mainstream for table imputation [136]. Recent studies have proposed a deep neural network-based imputation method, demonstrating higher imputation accuracy on large-scale datasets. In the context of the new era, with the rapid development of LLMs, table processing technology has encountered new opportunities and challenges. According to the latest research, LLMs can learn rich language and knowledge representations through pretraining, enabling them to execute data imputation tasks more efficiently. These models excel not only in unimodal tasks but also enhance the intelligence level of imputation tasks through multimodal data fusion and adaptive reasoning techniques. For instance, TableGPT has proven to handle missing data in complex tables

more effectively [87]. Despite the significant progress made in current table cleaning methods, issues such as reducing computational complexity, enhancing interpretability, and adapting to diverse data remain to be addressed.

Table Fact Verification. Table fact verification aims to validate whether the raised descriptions are entailed or refuted by the given tabular data. Compared with table QA, table fact verification involves more complex retrieval and reasoning processes. The descriptions may contain multiple implicit sub-facts, requiring comprehensive decomposition, validation and combination to obtain the final results. TabFact [3] is the most popular dataset that requires verifying claim sentences from a single Wikipedia table. INFOTABS [137] contains info-boxes from Wikipedia with short sentence statements with possible labels of NEI (*i.e.*, not enough info). FEVEROUS [52] requires revealing evidence from tables and affiliated texts possibly. In domain-specific scenarios, SEM-TAB-FACTS [138] and SciTab [53] utilize scientific tables, showing challenges in scientific compositional reasoning with expert knowledge. In addition, EHRCOn [139] involves consistency checking between unstructured clinical texts and structured medical database tables. Most LLM-based QA approaches [90; 96; 100; 112; 114] can well handle table fact verification evaluated on TabFact, showing their reasoning ability on common corpus and simple statements. However, current studies are not qualified in long-context scenarios, since the text paragraph requires fine-grained atomic decomposition, correlation validation and logical combination. In addition, for domain-specific scenarios, LLMs may lack expert knowledge to verify the claims accurately and reliably.

C. Table Content Retrieval

Retrieving relevant content from tables based on natural language descriptions has garnered substantial attention. Existing approaches can be typically divided into end-to-end methods and text-to-SQL methods.

End-to-end methods. End-to-End models primarily rely on deep learning techniques, aiming to directly feed structured data into end-to-end models to generate retrieval results. The STR [1] method proposes a semantic matching-based table retrieval approach that leverages multiple semantic representations and similarity measures to significantly improve table retrieval performance. Table2Vec [140] employs neural language modeling to embed table data into vector spaces and enhances row/column population and table retrieval tasks by incorporating embeddings of table elements. [141] approach uses BERT for table retrieval, combining features from prior research and jointly training them with BERT to achieve better results in handling complex query scenarios. GTR [142] introduces a graph-based table retrieval framework, using multi-granular graph representation learning to capture both table content and layout, thereby improving the model’s robustness and generalizability. CLTR [2] presents an end-to-end system that retrieves relevant tables and locates the correct cells within new benchmarks.

Text-to-SQL methods. Text-to-SQL models allow users to interact with table data stored in a database without the

need to learn the query language in depth. CodeS [143] is an open-source series of pre-trained language models specifically designed for text-to-SQL tasks, achieving good accuracy with smaller parameter sizes through incremental pre-training and bi-directional data augmentation. DTS-SQL [144] introduces a two-stage fine-tuning approach that enhances execution accuracy and closes the performance gap between small open-source models and large proprietary ones. DIN-SQL [145] improves the reasoning abilities of LLMs by decomposing the text-to-SQL task into smaller sub-tasks, achieving a significant accuracy boost on benchmarks like Spider and BIRD. CHESS [146] is a multi-agent framework that enhances text-to-SQL synthesis by addressing challenges in database schema navigation, query generation, and validation, offering privacy-preserving and scalable solutions.

D. Table Classification

Table classification tasks can be divided into table-level and column-level classification [147]. Table-level classification involves classifying the entire table based on its structure and content, while column-level classification focuses on classifying individual columns based on their data patterns. The key difference lies in granularity: table-level focuses on the whole table, and column-level focuses on specific columns.

Table-level Classification. In table-level classification, it is important to focus on the relationships among table structures, contents, metadata (e.g., headers or captions), and contextual information. TableTime [148] achieves optimal performance on table classification tasks using zero-shot reasoning methods through similar neighbor retrieval and multi-path reasoning methods. InstructTime [149] adopts a multimodal learning-to-generate paradigm by integrating pre-trained language models and time series discretization, achieving enhanced cross-domain transferability and superior classification performance across diverse datasets.

Column-level Classification. In column-level classification, it is important to focus on the column types, value distributions, semantic features, metadata (e.g., column headers), and inter-column relationships. TabLLM [131] is a method for few-shot classification of tabular data using LLMs. It serializes tabular data into natural language strings and adds task-related prompts to guide LLMs to make classification predictions. The UniPredict [150] method adopts a universal tabular modeling approach by leveraging generative LLMs, prompt engineering, and target augmentation, achieving state-of-the-art performance in tabular data classification and exceptional adaptability in few-shot learning scenarios. LIFT [130] converts non-language tasks into sentence templates and fine-tunes pre-trained language models without modifying their architecture, achieving strong performance in classification and regression.

E. Table-to-Text Generation

Table-to-text generation involves generating natural language descriptions or summaries from a given structured table of data [151; 152; 153]. The primary goal is to convert tabular data into a coherent and informative text that accurately

conveys the information contained in the table. The generated text should be fluent, informative, and effectively summarize the key insights from the table, including trends, patterns, comparisons and others [48; 56; 154]. The methods can be divided into the following categories: prompting-based methods, fine-tuning methods [155].

Prompting-based Methods. By designing appropriate prompts, the tabular data can be directly fed into the pre-trained LLMs for comprehensive processing. [156] investigates the generation capabilities of LLMs in real-world information-seeking scenarios, focusing on data insight generation and query-based generation. [155] explores the effectiveness of in-context learning strategies for table-to-text generation using open-source LLMs. The research evaluates zero-shot, single-shot, and few-shot prompting across benchmark datasets. [5] evaluates the performance of GPT-3.5 and LLaMa2-7B in the table-to-text generation task, exploring the effect of zero-shot prompts and fine-tuning using the LoRA method, and shows that GPT-3.5 performs well in the zero-shot setting. [157] proposes the TableGPT model, which achieves few-sample table-to-text generation through table structure reconstruction and content matching. The model uses prompting technology to convert table data into natural language descriptions. [158] introduces the TableGPT2 model, which combines tabular data with text and image data and uses hinting technology to achieve multimodal table-to-text generation. [156] investigates the application of LLMs in table-to-text generation for real-world information-seeking scenarios, proposing two new datasets and demonstrating the strong capabilities of LLMs in generating coherent, faithful narratives, assessing factual consistency and providing feedback to improve smaller fine-tuned models.

Fine-tuning Methods. To make LLMs better suited for scenario-specific tasks, most studies fine-tune the LLMs with specific datasets. This process requires labeled data to achieve improved generation quality. HeLM [159] performs parameter-efficient fine-tuning on the LLaMA2 model to improve the performance of table-to-text generation by emphasizing the data in specific rows of the table and injecting reasoning information into the input. [160] developed a table-to-text generation system by fine-tuning on a pre-trained large language model combined with a LoRA method to reduce the computational cost of fine-tuning. [5] evaluates the performance of GPT-3.5 and LLaMa2-7B on the table-to-text generation task, exploring the effects of zero-shot hints and fine-tuning using the LoRA method. [161] exploit the duality of generation and classification tasks and adopted an iterative generator-verifier approach to generate and verify training data from the language model to fine-tune a more powerful model that focuses on specific tasks without the need for manually labeled data.

The future of table-to-text generation lies in improving the factual consistency, contextual understanding, and adaptability of models across diverse domains [162]. Advancements in integrating structure-aware methods, enhancing multi-modal capabilities, leveraging LLMs with fine-tuning and prompt engineering, reducing hallucination and scaling to low-resource scenarios will be meaningful improvements.

F. Table Generation

Table generation involves converting unstructured or semi-structured data, like natural language descriptions, into organized tables that reflect data relationships and structure [163; 164]. The goal is to create coherent, consistent, and complete tables suitable for tasks such as data visualization, report generation, and knowledge representation [165; 166]. LLM-based methods for table generation are mainly divided into prompting-based and fine-tuning approaches.

Prompting-based Methods. The prompting-based method does not require additional fine-tuning of the model but relies heavily on the design of prompts to accurately guide the model in organizing the information into rows, columns, and cells, ensuring coherence and correctness of the generated tables. [167] examines the ability of several LLMs to generate factual and accurate tables using different prompting methods, including full-table, row-by-row, and cell-by-cell approaches. [168] presents an innovative auto-prompt generation system suitable for multiple LLMs, with minimal training. [169] evaluate the performances of state-of-the-art LLMs on this task in both fine-tuning and zero-shot settings, and additionally propose a novel pipeline called T3 to improve their performances. [170] further proposes TKGt, a two stages domain-aware pipeline. [4] apply prompted LLMs to this task, assessing their suitability as generalizable text-to-table systems.

Fine-tuning Methods. By fine-tuning pre-trained LLMs on specific datasets, LLMs learn to better align with the requirements of table generation tasks, such as structuring rows, columns, and cells, as well as preserving relationships within the data. This method enhances the model's ability to generate accurate and domain-specific tables but requires labeled training data and computational resources for fine-tuning. [171] presents TableLlama, an open-source generalist model for diverse table-based tasks, developed by fine-tuning Llama2-7B with LongLoRA, and introduces TableInstruct, a new dataset for instruction tuning and evaluating LLMs on table tasks. [161] introduces a generator-validator paradigm that iteratively generates and validates training data from language models to fine-tune specialized models for table tasks without requiring manually labeled data.

G. Table Manipulation

Table manipulation refers to the process of understanding, processing, and generating table data using machine learning and deep learning techniques. With the success of large models in natural language processing tasks, they have also shown great potential in the field of table manipulation. Table manipulation has emerged as a key area in enhancing the capabilities of LLMs for practical applications, particularly in handling spreadsheets and tabular data. SpreadsheetLLM [172] introduces a framework for encoding spreadsheets specifically for LLMs, enabling these models to better understand and manipulate spreadsheet data for a variety of tasks. TableLLM [173] builds on this by enabling real-world office scenarios where LLMs perform tabular data manipulation, offering a comprehensive approach to improving office productivity with table-related tasks. Furthermore, SheetAgent [174] presents

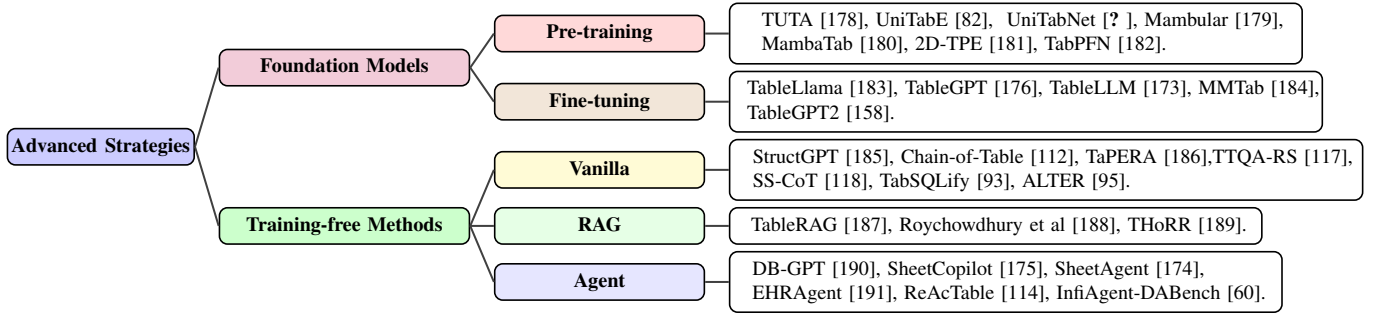


Fig. 8. Taxonomy and approaches of advanced LLM-based table mining.

a generalist agent that leverages LLMs for spreadsheet reasoning and manipulation, showcasing its ability to handle complex spreadsheet tasks in various domains. SheetCopilot [175] takes this a step further by focusing on software productivity, demonstrating how LLMs can assist in automating spreadsheet-related processes, enhancing workflow efficiency. TableGPT [176] aims to unify tables, natural language, and commands into a single GPT-based model, offering a versatile tool for integrating text and tabular data manipulation. MiMoTable [177] provides a multi-scale benchmark for spreadsheet reasoning, incorporating meta-operations to evaluate the reasoning abilities of models on tabular data. Together, these works demonstrate significant advancements in table manipulation, showcasing the potential of LLMs to automate, optimize, and transform the way we work with structured data. These advancements are driving intelligent spreadsheet management and bringing revolutionary solutions to data analysis and daily office tasks.

VII. ADVANCED LLM-BASED TABLE MINING

In the era of LLMs, novel learning paradigms have emerged in table mining tasks, achieving promising results in various scenarios. This section describes two advanced learning paradigms in LLM-based table mining, *i.e.*, table foundation models and training-free methods, whose overview is shown in Fig. 8).

A. Table Foundation Models

In the era of LLMs, universal table modeling represents a new paradigm that enables a single model or unified framework to address multiple tabular tasks, focusing on generalizing the understanding of table data within a single model architecture. It leverages accumulated general knowledge to achieve superior performance in complex tasks. To build a universal table foundation model, researchers must address challenges and opportunities across multiple dimensions, including handling table structures, row-column relationships, interactions between cells, and the integration of table data with other modalities. Table foundation models contain two mainstream strategies: pre-training and fine-tuning.

Pre-training. The pre-training method leverages large-scale, multi-source, and diverse tabular data (sometimes including related text descriptions, knowledge bases, or other modal data) to pre-train the model. This process enables

the model to learn general representations of table structures, data distributions, semantic information, and contextual dependencies. After acquiring these general representations, the model can flexibly transfer to a variety of table-related tasks. UniTabNet [?] combines vision and language models to propose a novel approach that enhances the recognition of table structures. This model leverages pre-trained vision-language multi-modal models to effectively capture complex structural information in tables, thereby improving automated table comprehension. Mambular [179], on the other hand, focuses on the application of sequential models in tabular deep learning. By pre-training to learn patterns in tabular data, it successfully enhances performance in table classification and regression tasks. This approach demonstrates the potential of pre-trained models in handling structured data. Additionally, MambaTab [180] introduces a plug-and-play model framework specifically designed for tabular data learning. By leveraging the advantages of pre-trained models, it shows flexibility and efficiency across a variety of tabular tasks. Furthermore, the TUTA framework [178] leverages tree-based transformers to better capture the structural nuances of tabular data, thereby enhancing scalability and interpretability. Building on this, 2D-TPE [181] further advances table understanding by applying 2D positional encodings, which effectively model the complex relationships between rows and columns. In a similar vein, UniTabE [82] addresses the diversity of tabular datasets by providing a unified encoder, offering a versatile and generalized solution for pre-training across various types of tabular data. TabPFN [182] is a transformer-based generative foundation model that is learned across millions of synthetic datasets, allowing fine-tuning, data generation, density estimation and learning reusable embeddings. These studies suggest that pre-training methods, by providing rich prior knowledge, can significantly improve the handling of tabular data, particularly in complex table understanding tasks.

Fine-tuning. The fine-tuning method builds upon existing pre-trained models, which is further trained or fine-tuned for the target application by leveraging task-specific or domain-specific data. Compared to training from scratch, this method allows for faster convergence to high accuracy while significantly reducing computational and data requirements. In office environments, TableLLM [173] enhances the practicality of table operations by fine-tuning pre-trained models on extensive datasets focused on table-related tasks. Similarly, TableLlama [183] develops a general-purpose model that can handle a

broad range of table-related tasks, achieved by fine-tuning on a large-scale table dataset. In the field of multi-modal table understanding [184], researchers have improved models' ability to comprehend complex tables by integrating data from multiple modalities—such as tables, text, and images—via fine-tuning. TableGPT [176], trained on multi-modal datasets that include tables, natural language, and commands, enables unified handling of natural language interactions with tables. Its successor, TableGPT2 [158], further enhances performance in multi-modal tasks by introducing a novel table encoder design. Collectively, these studies demonstrate the effectiveness of the fine-tuning approach in enhancing table model performance, providing efficient and adaptable solutions for a variety of tasks.

B. Training-free Methods

The training-free approach bypasses explicit pre-training or fine-tuning, instead leveraging the intrinsic capabilities of existing large models (e.g., GPT-series, BERT-derived models, multi-modal models, or domain-specific models) to handle table-related tasks. By carefully crafting prompts and input formats, table information will be reformulated into representations that the model can easily process. Essentially, the training-free method operates within the prompting paradigm. Associated studies include vanilla prompting, retrieval-augmented generation (RAG), and agent-based models.

Vanilla Prompting. In recent years, significant progress has been made in table understanding and question-answering tasks, with many studies focusing on enhancing the reasoning capabilities of LLMs through the design of sophisticated prompts. For example, Chain-of-Table [112] introduces a strategy that evolves the table within the reasoning chain, thereby improving the depth and accuracy of table understanding. TaPERA [186] enhances the reliability and interpretability of long-text table question answering through content planning and executable reasoning. Meanwhile, StructGPT [185] offers a general framework that helps LLMs effectively handle structured data, boosting their reasoning abilities via advanced prompt design. In the domain of multi-hop table-text question answering, TTQA-RS [117] proposes a decomposition-based multi-hop reasoning method, effectively improving both the precision and conciseness of the reasoning process. SS-CoT [118] further explores enhancing table question answering by refining reasoning mechanisms and optimizing model performance in complex tasks. TabSQLify [93] strengthens the reasoning capabilities of LLMs through table decomposition, while Lastly, ALTER [95] proposes an enhancement technique to boost the reasoning capabilities of LLMs when handling large tables, offering a fresh perspective on table understanding.

RAG. Retrieval-augmented generation (RAG) aims to improve reasoning quality by retrieving highly related information to support the understanding and reasoning processes. In table mining, numerous RAG-based studies achieve promising results. TableRAG [187] leverages millions of tokens to retrieve and generate table information, greatly improving

question-answering performance. In a case study on 3GPP specifications, [188] evaluates various table representation methods, underscoring the RAG model's ability to address table-related queries within complex documents. Additionally, THoRR [189] advances RAG by optimizing generation quality through refined table retrieval and enhancement techniques, ensuring higher accuracy and consistency for intricate table tasks. These studies demonstrate that RAG significantly boosts table mining performance by integrating external knowledge bases and retrieval capabilities, especially when dealing with large-scale table data.

Agent. Recently, LLM-based agents have made significant advancements in table reasoning and database interaction. ReAcTable [114] refines the ReAct method to improve response and action strategies specifically for table QA tasks. SheetAgent [174] introduces a versatile agent that leverages LLMs to assist with spreadsheet data processing and reasoning tasks. EHRAgent [191] integrates coding capabilities to enable few-shot learning, effectively managing complex table reasoning tasks in electronic health records. In the domain of data analysis, InfiAgent-DABench [60] provides a standardized benchmark for evaluating intelligent agents across various tasks, contributing to the progress of research in this area. DB-GPT [190] enhances database interactions by utilizing proprietary LLMs, enabling users to engage with databases more naturally and efficiently. Additionally, Sheet-Copilot [175] integrates intelligent agents with productivity software tools, improving the efficiency of table processing and data analysis, while showcasing the potential of LLMs in office applications. These studies highlight the widespread application of LLM-based agents, leading to the potential development of automated and intelligent table-related office tools.

VIII. TRUSTWORTHY LLM-BASED TABLE MINING

Given the black-box nature of LLMs, it is crucial to make their intermediate reasoning process traceable and easy to understand, especially in applications in vertical fields (e.g., healthcare, finance). Trustworthy table mining requires ensuring that the mined knowledge and discovered patterns are credible, reliable, and can be effectively verified and explained during the analysis of tabular data. In trustworthy table mining, several aspects need to be addressed.

Robustness. Robustness refers to the model's ability to make effective decisions stably and reliably in the face of uncertainties (e.g., noise, outliers, missing data, distribution changes) in the data [103; 105; 192]. Some tabular studies have addressed the robustness against human-annotated adversarial perturbations [103] and attempted to explore the essential factors towards fine-grained robustness [105]. Currently, how to balance robustness and model complexity, how to deal with local anomalies or distribution deviations in large-scale data, and how to ensure the consistency and reliability of the model in the case of multi-source heterogeneous data still remain challenges.

Explainability. Explainability refers to making the reasoning process, mechanism, and results transparent for precise

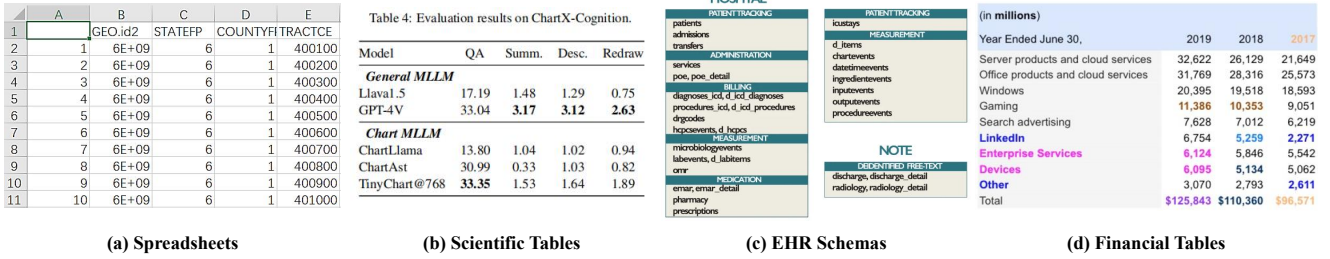


Fig. 9. Domain-specific applications in (a) spreadsheets, (b) scientific literature, (c) electronic health records and (d) finance.

comprehension of how models make decisions based on the input data [106; 111; 120]. By revealing precise intermediate steps, some LLM-based studies have improved the explainability of the produced answers [111; 120]. However, current approaches still face challenges, including the contradiction between complex models and explainability, the balance between accuracy and explainability, and the compositional path presentation in heterogeneous table processing.

Privacy Preserving. Privacy preserving refers to taking effective measures to ensure that sensitive information is not leaked or abused and to protect individual privacy rights when analyzing and mining tabular data, especially in sensitive fields (e.g., healthcare, finance) [193; 194; 195]. Existing challenges include how to ensure data availability and accuracy of analysis results without leaking privacy, how to balance privacy protection and computational efficiency, and how to deal with privacy conflicts when sharing data across institutions or multiple parties.

IX. DOMAIN-SPECIFIC TABLE MINING

With the continuous growth of tabular data and the development of artificial intelligence technology, various fields have gradually driven the demand for automated table mining. This section introduces some representative fields of intelligent understanding and application scenarios of tabular data, covering spreadsheets, scientific literature, electronic health records and financial statements. Examples of the four types of domain tables are shown in Fig. 9.

A. Spreadsheet Tables

Spreadsheets cover flexible table forms and types so that they are widely utilized to design table understanding and reasoning assessments from scenarios of daily work. Numerous studies extract spreadsheet tables to build a variety of datasets addressing different challenges. HiTab [48] is a cross-domain QA dataset containing numerous hierarchical tables, which are in spreadsheet forms for easy writing and checking. ForTap [196] first explores leveraging spreadsheet tables for table pre-training under two numerical-aware self-supervised pre-training objectives. SpreadsheetCoder [197] is a BERT-based architecture for synthesizing spreadsheet formulas from tabular contexts, containing headers and table contents. SheetCopilot [175] is an agent that takes users' daily tabular processing tasks in natural language to fulfill the requirements, in which a state machine-based task planning framework is designed

for LLMs to robustly interact with spreadsheets. SpreadsheetBench [59] is a challenging spreadsheet manipulation benchmark built from real questions gathered from online Excel forums, incorporating a reliable evaluation metric akin to online judge platforms. MiMoTable [177] is a multi-scale spreadsheet benchmark with meta-operations for table reasoning, covering seven domains and different types of real-world spreadsheets with a new difficulty measuring criterion.

B. Scientific Tables

The field of scientific research produces an increasing amount of scientific documents (e.g., literature, patents). Nowadays, scientific mining has become a crucial research issue that attracts substantial interest [198; 199; 200]. Based on the huge demand of scientific researchers for reading, understanding and analyzing scientific documents, recent years have witnessed a rising trend of studies on automatic content retrieval [201; 202], multi-modal chart understanding [203; 204] and domain-specific knowledge extraction [205; 206]. Among these issues, scientific tables are widely used as an important information carrier to display core key information (e.g., experimental results, chemical reaction conditions). Automated scientific table understanding and mining is also a popular research topic [21]. Earlier studies focus on table structure and content recognition. SciTSR [37] and PubTables-1M [40] are the specified scientific table recognition datasets, including various types of tables from huge amounts of scientific literature in the PDF format. Research methods for scientific table structure recognition have gradually matured in recent years. Since the rise of pre-trained language models and LLMs, scientific table understanding and reasoning have become an increasingly important research focus in academia. S2abEL [57] is the first dataset for scientific table entity linking, whose challenge lies in the incompleteness of the scientific knowledge base and the fact that the elimination of related ambiguities usually requires understanding the text of scientific papers as well as tables. SciTab [53] is a scientific table fact-checking dataset that overcomes the bias caused by crowdsourced claims and over-reliance on text-based evidence. ArxivDIGESTables [207] contains literature review tables extracted from ArXiv research papers. SciTabQA [208] is a scientific table QA benchmark dataset that can be used to study question answering on scientific heterogeneous data and has strong robustness.

C. Electronic Health Record Tables

The healthcare industry has become increasingly digital and computerized [209]. In hospitals and laboratories, daily-generated patient records and data of clinical trials are countless stored in specific systems, known as electronic health record (EHR) [210; 211] systems involving complicated structured databases and unstructured texts. Based on public databases (e.g., MIMIC [212; 213], eICU [214]) and non-public hospital de-identified real datasets, existing studies have provided numerous insights into finding clinical evidence and medical decision-making. EHR data possess several characteristics: huge volume, complex relationships, rich semantics, and diverse modalities, indicating that a full understanding of them may require a lot of energy from overburdened medical workers. Therefore, automated retrieval and understanding of EHR data are of great significance for medical workers. emrQA [215] addresses the lack of publicly available EHR QA corpus, re-purposing existing annotations to build a large-scale EHR QA dataset. MIMICSQL [216] creates a novel large-scale Question-to-SQL pair dataset based on public MIMIC EHR data, addressing the issue of effective and efficient patient information retrieval from EHR data. MIMICSQL* and MIMIC-SPARQL* [217] are the modified versions of MIMICSQL allowing KG-based EHR QA through database relations in the form of directed acyclic graph. EHRSQL [218] is an EHR text-to-SQL benchmark that reflects a wide range of needs in the hospital, covering more than 10 tables and a variety of real-world questions. DrugEHRQA [219] is a QA dataset containing medication-related queries, constructed from both structured tables and unstructured notes. EHRXQA [220] is a multi-modal QA dataset combining structured EHRs and chest X-ray images, necessitating both uni-modal and cross-modal reasoning. EHRCon [139] is a dataset about checking the consistency between unstructured notes and structured EHR tables since unintentional errors may be recorded into EHRs, along with a dedicated workflow to reach the checking goal. EHRAgent [191] is an LLM-based agent that enables autonomous code generation and execution to facilitate clinical interactions.

D. Financial Tables

The fields of business and finance are full of statistical and financial reports. Companies may use market trend information in statistical reports to predict future economic performance and adjust their financial plans accordingly. This type of mining is of great significance for investor decision-making, risk assessment, compliance checking, and corporate performance analysis. Current approaches have addressed the issues of numerical reasoning, special formats and others in financial table mining. TAT-QA [47] is a QA benchmark on a hybrid of tabular and textual content, which are extracted from real financial reports including numerical reasoning within common operations. FinQA [46] is another QA dataset containing QA pairs over financial reports written by domain experts. AIT-QA [221] is a QA dataset consisting of airline industry tables that address more complex layouts. MultiHiertt [50] addresses multi-hierarchical tabular and textual data in real financial

reports, which raises challenges towards the existing baselines. ConvFinQA [222] is a QA dataset that aims to study the chain of numerical reasoning in conversational QA in financial scenarios. SoarGraph [223] uses hierarchical graphs to facilitate supporting evidence extraction and enhance numerical reasoning in finance table-text understanding. FinanceMATH [224] is a novel benchmark for LLMs' capability evaluation, containing a hybrid of table-text problems that require college-level knowledge and a knowledge bank in the finance domain. [225] conduct an evaluation study over several financial table QA datasets that focuses on sensitivity to table complexity and changes in performance as the number of arithmetic reasoning steps increases.

X. PROSPECTS AND FUTURE DIRECTIONS

In this section, we outline some prospects and future directions about the field of LLM-based table mining, which we believe are critical to further potential advancements.

Long-context Scalability. Most scenarios and applications involve large-scale tables or lengthy texts that require long-context processing. Although LLMs that can handle million tokens as input have emerged, they still struggle with complicated long-context mining tasks, suffering from poor long-term memorization and high computational costs. Efficient evidence retrieving, decomposed understanding procedures, and specific memory mechanism designs can be considered improvements in the accuracy and efficiency of long-context table mining.

Multi-modal Content Understanding. Tables can store multi-modal elements including numerical values, texts, images and even videos. Existing research has widely explored studies on understanding numerical values and textual contents in text-form or visual-form tables, yet cell contents of images, videos and other modalities have been underexplored. Through modality-specific modules, it may be easy to understand them separately. However, how to align these cells in different modalities under structured table semantics remains challenging.

Real-time Interactive Processing. Most real-world applications require real-time interactive table mining processes, such as spreadsheet manipulation, report table generation. Uncertainty exists in user queries since they may differ in styles, difficulties, scenarios and others. How to precisely capture users' intentions and finely adapt them to the real scenarios from interactive responses can be considered in future studies.

Agentic Reasoning Paradigm. There have been numerous LLM-based table mining frameworks equipped with hand-crafted prompts and structured workflows. Although they demonstrate promising results in some specific tasks and applications, they fall short in generalization since they show little ability in autonomous learning, planning and executing in general scenarios. Therefore, developing the agentic reasoning paradigm in table mining can be another cutting-edge issue in LLM-based table mining.

Trustworthy Table Mining. Existing LLM-based table mining studies have achieved promising accuracy in most tasks, yet they focus less on trustworthy issues, including

explainability, robustness, fairness and privacy preservation. Some studies have investigated trustworthy improvements in general scenarios, while in some domain-specific applications (e.g., medical diagnosis, financial analysis), trustworthy issues have been less regarded. Research in trustworthy table mining will be significant and popular for a long time.

Universal Table Foundation Models. Foundation models have been widely explored in domains such as languages, visions, robots and others. They require large-scale training within heterogeneous data and various tasks and can adapt to downstream applications in a fast and efficient manner. While some studies have explored such issue, there remains a long way to achieve real universal table foundation models since they require fine-grained module designs and huge training costs to cover a wide range of table-based tasks. Despite its huge difficulty, developing universal table foundation models are quite meaningful.

Domain-specific Knowledge Integration. Domain-specific tables and their affiliated information can store key concepts and terminologies, which are difficult to understand through general-purpose LLMs. For example, medical EHR tables and notes commonly include medical terminology, special acronyms and non-grammatical shorthand. Therefore, how to realize effective and efficient knowledge integration in some crucial domains is another challenging future direction, as current studies have not deeply explored this issue.

By addressing these challenges and opportunities, we believe that the related studies will significantly advance LLM-based table mining, which will continue to be an energetic and meaningful research topic for a long period.

XI. CONCLUSION

In this paper, we conducted a systematic and extensive review of the research on table mining with large language models (LLMs). We thoroughly overviewed and analyzed the concepts and unique challenges posed by tabular data and its integration of LLMs. Then, we reviewed and compared a wide range of LLM-based approaches for various table preparation and mining tasks, highlighting critical advancements in contextual embedding, reasoning capabilities, and cross-modal integration. Next, we described advanced LLM-based learning strategies, including foundation models and training-free methods. Additionally, we presented trustworthy techniques integrated into table mining studies, followed by domain-specific table mining applications. Finally, we discussed some of the most pressing open challenges and future research directions, such as enhancing model generalization, improving interpretability, and addressing computational efficiency. As this survey provides a comprehensive view of the broader challenges, advancements and application scenarios in table mining, we hope that it provides readers with a valuable resource for understanding and clustering current advanced methods and techniques, inspiring novel prospects and notable progress in future studies.

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